AID-DATA-005

PRODUCT IDENTIFICATION & VARIANT CODING STANDARD

Final coding architecture for scalable product, variant and multi-seller data in AIDOOi

1. Purpose

This standard defines the final AIDOOi identification architecture for product families, physical product variants and seller-specific offers. The objective is to keep identifiers stable, globally unique and independent from taxonomy, while allowing AIDOOi AI to use clean structured product data for matching.

2. Final Architecture

Record Type	AIDOOi Code Format	Purpose
Parent Product	AIP-00000001	Groups variants belonging to the same underlying model or product family.
Product / Variant	AID-00000001	Identifies one canonical physical product variant in the AIDOOi catalog.
Seller Offer	AIO-00000001	Identifies one seller-specific commercial offer for a Product, including price, stock and seller SKU.
Seller SKU	Seller supplied	Seller's own inventory/commercial code; stored exactly as supplied.

Why no category abbreviation inside Product ID?

The Product ID must identify the record, not describe it. Category and Product Type can be corrected or expanded later, while the Product ID must never change. AIDOOi AI must read Main Category, Category, Product Type and structured attributes directly rather than infer meaning from an ID string.

3. Numbering Rules

Rule	Standard
Prefix	AIP = Parent Product; AID = Product; AIO = Seller Offer.
Numeric block	Eight-digit global sequential number, zero-padded.
Sequence	Each record type uses its own global sequence.
Uniqueness	Every code is unique within its record type and is never reused.
Permanence	Once assigned, the code never changes, even if category, price, stock, seller or descriptive data changes.
Allowed characters	Uppercase English letters, numbers and hyphens only.
Meaning inside ID	Do not encode category, colour, material, seller, location, price, stock or other mutable data.

4. Real Example - The AIDOOi Test Chair

Assume the same lounge-chair model is available in Brown, Beige and Black. The three colour options are variants of the same underlying model.

Record	Code	Relationship / Variant
Parent Product	AIP-00000001	Underlying chair model
Brown Product	AID-00000001	Parent = AIP-00000001; Colour = Brown
Beige Product	AID-00000002	Parent = AIP-00000001; Colour = Beige
Black Product	AID-00000003	Parent = AIP-00000001; Colour = Black

Taxonomy for all three variants:

Field	Value
Main Category	Seating
Category	Chair
Product Type	Lounge Chair

5. Why Seller Offer Needs Its Own ID

AIDOOi may aggregate multiple retailers. The same canonical Product can therefore be sold by more than one seller with different SKUs, prices, stock levels, showroom locations or delivery conditions. Those commercial differences must not create duplicate Product identities.

Seller Offer	Product ID	Seller	Seller SKU	Price / Stock
AIO-00000001	AID-00000001	Blossom Home	RASO_D-6604	Seller-specific
AIO-00000002	AID-00000001	Another Retailer	CHAIR-BR-91	Seller-specific

Both offers point to the same Brown chair Product. AIDOOi can therefore recommend the product once, then choose or display the best available seller offer based on price, availability, delivery and other commercial criteria.

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6. Variant Decision Rule

Same underlying model + customer-selectable difference = same Parent Product.
Different fundamental model, structure or function = new Parent Product.

Difference	Normally Same Parent?	Example
Colour	Yes	Brown / Beige / Black
Upholstery / Fabric	Yes	Fabric A / Fabric B
Finish	Yes	Oak / Walnut finish
Size	Conditional	Same model offered in approved size options
Orientation	Usually Yes	Left-facing / Right-facing sectional
Configuration	Conditional	Same commercial model with selectable configuration
Fundamental design change	No	Different model identity
Major structural/function change	Usually No	Standard sofa vs sofa bed unless manufacturer treats them as one model family

7. Product Without Variants

If a product has no genuine variants, it receives a Product ID directly and **Parent Product ID = NULL**. Do not create artificial Parent records only to satisfy the data structure.

Field	Example
Product ID	AID-00000020
Parent Product ID	NULL
Main Category	Tables
Category	Living Room Table
Product Type	Coffee Table

8. Example - Entering a Table After the Chair

Product numbers are global and continue sequentially regardless of category. If the first three Product IDs are the chair variants above, the next new standalone product can simply be:

Field	Coffee Table Example
Product ID	AID-00000004
Parent Product ID	NULL
Main Category	Tables

Field	Coffee Table Example
Category	Living Room Table
Product Type	Coffee Table

The number 00000004 does not need to indicate that the item is a table. The structured taxonomy fields provide that information explicitly and reliably.

9. Optional Human-Readable Classification Code

For operational screens, spreadsheets or warehouse-facing reports, AIDOOi may display a separate controlled classification code such as **CHR**, **TBL** or **LGT**. This code is not part of the immutable Product ID and must not be used by the AI as the source of taxonomy truth.

System Product ID	Classification Code	Displayed Together If Useful
AID-00000001	CHR	CHR | AID-00000001
AID-00000004	TBL	TBL | AID-00000004
AID-00000025	LGT	LGT | AID-00000025

10. AI and Matching Rule

AIDOOi AI must never parse Product ID, Parent Product ID, Offer ID or Seller SKU to understand the product. Matching must use explicit structured fields including Main Category, Category, Product Type, dimensions, colour, material, style, price, availability and category-specific attributes.

11. Lifecycle Rules

Situation	Required Action
Product goes temporarily out of stock	Keep Product ID unchanged; update seller Offer stock status.
Seller discontinues its listing	Keep Product ID; mark that Seller Offer discontinued/inactive.
Another seller offers the same exact variant	Create a new AIO Offer ID linked to the existing Product ID.
New colour of same model appears	Create a new Product ID linked to the same Parent Product ID.
Product taxonomy is corrected	Update taxonomy fields; do not change Product ID.
Product is permanently retired	Keep historical IDs; mark relevant status as discontinued/archived. Never reuse IDs.

12. Migration / Supersession Note

This standard supersedes the earlier category-embedded Product ID format such as **AI-CHR-000001** and the corresponding Parent format such as **AI-CHR-P000001**. The conceptual Parent/Variant rules remain valid, but identifier formatting must follow AID-DATA-005 going forward.

Final Standard

Parent Product: AIP-00000001
Canonical Product / Variant: AID-00000001
Seller Offer: AIO-00000001
Seller SKU: stored separately exactly as supplied.

Taxonomy and product intelligence remain separate structured data and are never encoded into the permanent identifier.